

Extremes for metocean: Recent developments and take-aways

Philip Jonathan

Lancaster University

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(slides at www.lancs.ac.uk/~jonathan)



Overview of masterclass

- A: Fundamentals
- B: Covariates
- C: Multivariate
- D: covXtreme
- E: Recent developments and take-aways

Overview of E: Recent developments and take-aways

Recent developments

- Full probabilistic design
- Model of within-storm dependence of sea states
- Angular-radial model for multivariate extremes
- Return values with uncertainty

Take-aways

- Key things to remember
- Best practice mantra

Full probabilistic design

Set-up

- Storm storm peak events X^{sp} dependent on covariates Θ^{sp}
- An evolving within-storm environment $\{(\mathbf{X}_s, \Theta_s)\}_{s \in S_T}$ for storm of random length T
- Structural “loading” $\mathbf{Y}$
- Everything subject to sources of uncertainty $\mathbf{Z}$
- $\mathbf{Z}, \Theta^{\text{sp}}, X^{\text{sp}}, \{(\mathbf{X}_s, \Theta_s)\}_{s \in S_T}$ and $\mathbf{Y}$ are multidimensional

Full probabilistic design

Unconditional distribution of loading for a random storm

$$\begin{aligned} F_Y(\mathbf{y}) &= \int_{\zeta} \int_{(\{\mathbf{x}_s, \boldsymbol{\theta}_s\}_{s \in \mathcal{S}_T}, \tau)} \int_{\mathbf{x}^{\text{sp}}} \int_{\boldsymbol{\theta}^{\text{sp}}} \\ &\times F_{Y|\{\{\mathbf{x}_s, \boldsymbol{\theta}_s\}_{s \in \mathcal{S}_T}, \mathbf{Z}(\mathbf{y}|\{\{\mathbf{x}_s, \boldsymbol{\theta}_s\}_{s \in \mathcal{S}_T}, \zeta)} \\ &\times f_{(\{\{\mathbf{x}_s, \boldsymbol{\theta}_s\}_{s \in \mathcal{S}_T}, T)|\mathbf{x}^{\text{sp}}, \boldsymbol{\theta}^{\text{sp}}, \mathbf{Z}} \left(\{\{\mathbf{x}_s, \boldsymbol{\theta}_s\}_{s \in \mathcal{S}_T}, \tau \mid \mathbf{x}^{\text{sp}}, \boldsymbol{\theta}^{\text{sp}}, \zeta \right) \\ &\times f_{\mathbf{x}^{\text{sp}}|\boldsymbol{\theta}^{\text{sp}}, \mathbf{Z}}(\mathbf{x}^{\text{sp}}|\boldsymbol{\theta}^{\text{sp}}, \zeta) \\ &\times f_{\boldsymbol{\theta}^{\text{sp}}|\mathbf{Z}}(\boldsymbol{\theta}^{\text{sp}}|\zeta) \\ &\times f_{\mathbf{Z}}(\zeta) \\ &\times d\boldsymbol{\theta}^{\text{sp}} d\mathbf{x}^{\text{sp}} d(\{\{\mathbf{x}_s, \boldsymbol{\theta}_s\}_{s \in \mathcal{S}_T}, \tau) d\zeta \end{aligned}$$

Full probabilistic design

Distribution of annual maximum loading

- For univariate load Y here

$$F_A(y) = \int_c [F_Y(y)]^c f_C(c) dc$$

- Density f_C of annual rate of occurrence of storms
- Return value for return period P years given by $F_A^{-1}(1 - 1/P)$

Expected annual utility for independent storms

- For multivariate load Y here

$$\mathbb{E}(U|\mathcal{R}) = \int_y U(y|\mathcal{R}) f_A(y) dy$$

- System utility $U(y|\mathcal{R})$ given system “strength” characteristics $\mathcal{R}$
- Solve for $\mathcal{R}$ to achieve required expected annual utility

Full probabilistic design

Expected annual utility for dependent storms

- For year with M random storms, M random

$$\begin{aligned}\mathbb{E}(U_A|\mathcal{R}) &= \int_m \int_{\mathbf{y}_1} \dots \int_{\mathbf{y}_m} U_A(\mathbf{y}_1, \dots, \mathbf{y}_m|\mathcal{R}) \\ &\times f_{Y_1, \dots, Y_m, M}(\mathbf{y}_1, \dots, \mathbf{y}_m, m) \\ &\times d(\mathbf{y}_1, \dots, \mathbf{y}_m, m)\end{aligned}$$

- System annual utility $U_A(Y_1, \dots, Y_m|\mathcal{R})$ given system “strength” characteristics $\mathcal{R}$
- $f_{Y_1, \dots, Y_m, M}$ is the joint density of multivariate loading from M random storms
- Solve for $\mathcal{R}$ to achieve required expected annual utility

Full probabilistic design

The quantities we need to estimate F_Y

- f_Z : a means of quantifying uncertainty; random thresholds, bootstrap resampling
- $f_{\Theta^{\text{sp}}|Z}$: a model for rate of occurrence of storms with covariates
- $f_{\mathbf{X}^{\text{sp}}|\Theta^{\text{sp}},Z}$: a model for storm peaks
- $f(\{(\mathbf{x}_s, \Theta_s)\}_{s \in S_T, T})|\mathbf{X}^{\text{sp}}, \Theta^{\text{sp}}, Z$: a model for within storm evolution
- We can estimate all these using the statistical models discussed here
- See Speers et al. [2024], Speers et al. [2026]

Model of within-storm dependence of sea states

Canonical extensions of the conditional extremes model

- Basic: $X|(Y = y), y > u$
- Temporal: “storm evolution” $X_1, X_2, \dots, X_\tau|(X_0 = x_0), x_0 > u$
- Spatial: “spatial dependence” $X_1, X_2, \dots, X_s|(X_0 = x_0), x_0 > u$

Idea

$$X_1, X_2, \dots, X_p|(Y = y) = \alpha y + y^\beta \mathbf{Z}$$

- Impose appropriate structure on parameters α, β and distribution of $\mathbf{Z}$
 - e.g. α evolves smoothly in time
 - e.g. $\mathbf{Z}$ follows a multivariate Gaussian or extension thereof with appropriate mean and covariance forms
- Make a simplifying assumption
 - e.g. $X_{t+1}, X_{t+2}|(X_t = x) = [\alpha_1 \alpha_2]x + x^{[\beta_1 \beta_2]}[Z_1 Z_2]$

Further extensions

- Non-stationary and multivariate temporal and spatial models

Model of within-storm dependence of sea states

Applied conditional extremes references

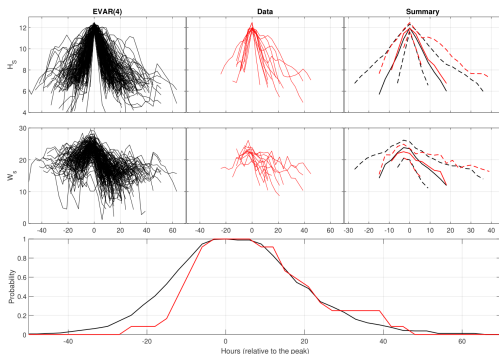
- Non-stationary : Jonathan et al. [2014]
- Time-series : Winter and Tawn [2016], Tendijsck et al. [2019], Tendijsck et al. [2024a]
- Mixture model : Tendijsck et al. [2023]
- Spatial : Shooter et al. [2021b], Shooter et al. [2021a], Shooter et al. [2022]

Model of within-storm dependence of sea states

Extremal vector auto-regression (EVAR; Tendijck et al. 2024a)

- On Laplace margins, with component-wise operations and $\mathbf{X}_t \in \mathbb{R}^d$:

$$\mathbf{X}_{t+k} | (\mathbf{X}_t, \dots, \mathbf{X}_{t+k-1}, X_{t,1} = y) = \sum_{\ell=1}^k A_{\ell} \mathbf{X}_{t+k-\ell} + y^b \mathbf{Z}, \quad y > u \gg 0$$



Excursions of H_S (top) and W_S (middle) from EVAR(4) model (left; black), observed (middle; red) on original margins with storm peak $H_S \in [11.5, 12.5]$; right-hand plots summarise the observed (red) and EVAR(4) (black) excursions, using median (solid), 10% and 90% quantiles (dashed). In the bottom panel, we plot survival probabilities for observed (red) and EVAR(4) (black) excursions relative to the time of the excursion maximum.

Angular-radial model for multivariate extremes

Semi-parametric angular-radial (SPAR) model

- Radial R and angular Q components. Then joint density factorised as

$$f_{R,Q}(r, q) = f_Q(q) f_{R|Q}(r|q)$$

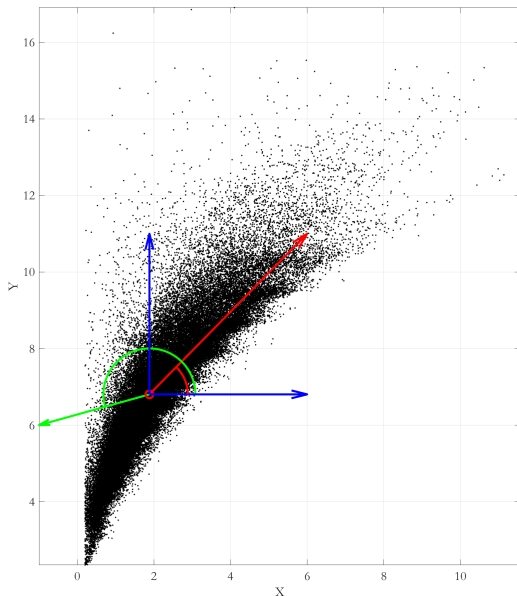
- Assume GP conditional tail for $R|(Q = q)$, with parameters varying smoothly with angle q above some threshold $\psi(q)$ with non-exceedance probability $\tau(q)$

$$f_{R,Q}(r, q) = f_Q(q) \times \tau(q) f_{GP}(r - \psi(q) | \xi(q), \sigma(q)), \quad r > \psi(q)$$

with smoothly varying $\psi(q)$, $\tau(q)$, $\xi(q)$ and $\sigma(q)$. Also assume angular density $f_Q(q)$ varies smoothly with q

- SPAR representation shown to provide good approximations to a large set of copula functions on standard margins
- Is transformation to standard margins necessary?
- Different possible angular-radial decompositions using “generalised co-ordinates”
- $\Rightarrow$ multivariate extremes is just “non-stationary univariate” extremes

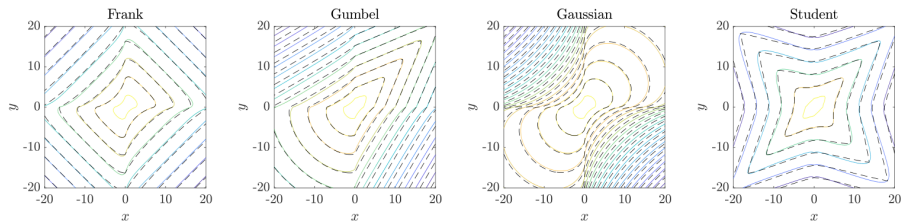
Angular-radial model for multivariate extremes



- SPAR 2-D
- $H_S(X)$ and $T_P(Y)$
- Radial and angular (R, Q) instead of Cartesian (X, Y)
- OK for all $Q \in [0, 360)$
- OK in higher dimensions
- Algorithms in 2-D same as for directional extremes!

Angular-radial model for multivariate extremes

SPAR fits to extreme value copulas

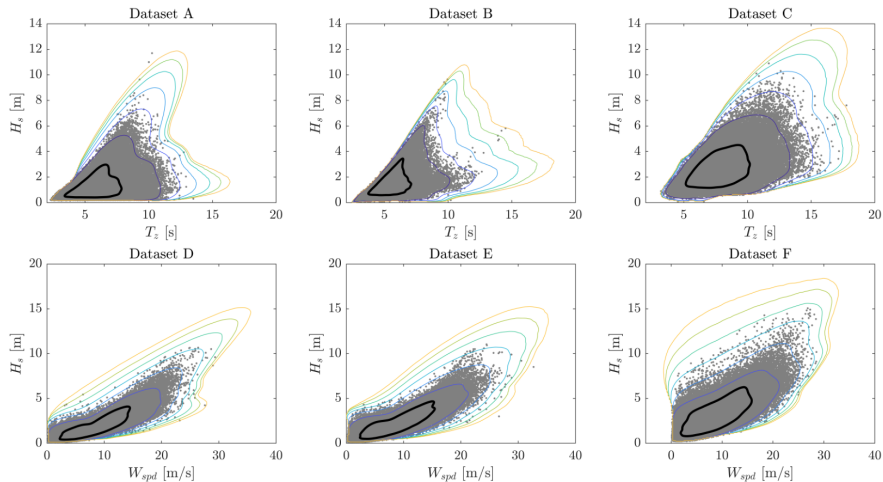


Density contours of various copulas on Laplace margins. All copulas have Pearson correlation coefficient 0.6. Student-t copula has two degrees of freedom. Solid lines: true contours at logarithmic increments. Dashed lines: SPAR-estimated contours.

- SPAR admits AI (e.g. upper tails of Frank and Gaussian) and AD (e.g. upper tails of Gumbel and Student-t)
- SPAR handles all directions (not just “first quadrant”)
- Link to limit sets (see Multivariate slides)

Angular-radial model for multivariate extremes

Density contours from SPAR fits to data



Density contours from SPAR model for 6 samples.

Angular-radial model for multivariate extremes

SPAR references

- Mackay and Jonathan [2023] (original work on arXiv)
- Murphy-Barltrop et al. [2025] (first peer-reviewed publication)
- Mackay et al. [2025] (5-D SPAR with deep learning)
- Parallel developments in the statistics literature

Return values with uncertainty

What is a return value?

- Random variable A represents the maximum value of some physical quantity X per annum
- Forget about all complicating issues like serial dependence, covariates and other sources of dependence and uncertainty
- The P -year return value x_P of X is then defined by the equation

$$F_A(x_P) = \Pr(A \leq x_P) = 1 - \frac{1}{P}$$

- Or

$$x_P = F_A^{-1}\left(1 - \frac{1}{P}\right)$$

- Typically $P \in [10^2, 10^8]$ years

Return values with uncertainty

An alternative definition

- Random variable A_P represents the P -year maximum value of X
- The P -year return value x'_P of X can be found from F_{A_P} for large P , assuming independent annual maxima since

$$\begin{aligned} F_A(x_P) &= 1 - \frac{1}{P} \\ \Rightarrow F_{A_P}(x_P) &= \left(1 - \frac{1}{P}\right)^P \approx \exp(-1) \end{aligned}$$

- Use $F_{A_P}(x'_P) = \exp(-1)$ to define an alternative return value x'_P

Return values with uncertainty

Estimating a return value

- To estimate x_p , we need knowledge of the distribution function F_A of the annual maximum
- We might estimate F_A using extreme value analysis on a sample of independent observations of A
- Typically more efficient to estimate the distribution $F_{X|X>u}$ of threshold exceedances of X above some high threshold u using a sample of independent observations of X , and use this in turn to estimate F_A and x_p
- For threshold u , estimate GP parameter ξ , σ , Poisson annual rate of occurrence of exceedances λ
- So what's the problem?

Return values with uncertainty

Parameter uncertainty

- x_P can be estimated easily in the absence of uncertainty
- In reality, we estimate parameters λ , ψ , σ and ξ from a sample of data, and we cannot know their values exactly
- How does this epistemic uncertainty affect return value estimates?
- A number of different plausible estimators for return values under uncertainty
- Different estimators perform differently (bias and variance)
- Which estimators are likely to perform reasonably in fairly general circumstances?
- Is it even sensible or desirable to estimate return values?

Return values with uncertainty

Predictive distributions

- If a distribution $F_{Y|Z}$ of random variable Y is known conditional on random variables Z , and the joint density f_Z of Z is also known, the unconditional predictive distribution $\tilde{F}_Y$ can be evaluated using

$$\tilde{F}_Y(y) = \int_{\zeta} F_{Y|Z}(x|\zeta) f_Z(\zeta) d\zeta$$

- The expected value of deterministic function g of parameters Z given joint density f_Z is

$$E[g(Z)] = \int_{\zeta} g(\zeta) f_Z(\zeta) d\zeta$$

- $\zeta = (\lambda, \psi, \sigma, \xi)$, $Y = A$ (or $Y = A_P$)
- In a Bayesian setting, $f_Z(\zeta)$ can be interpreted as the posterior distribution ($\triangleq f_{Z|D}(\zeta)$) of model parameters following a Bayesian inference, influenced by choice of (subjective) prior

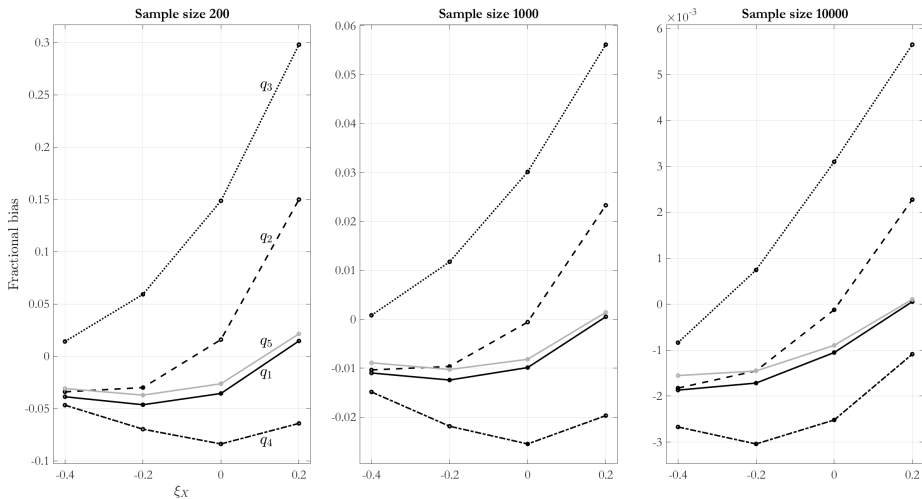
Return values with uncertainty

Different estimators of return value

$$\begin{aligned}q_1 &= F_{A|Z}^{-1}(1 - 1/P \mid \mathbb{E}_Z[Z]) = F_{A|Z}^{-1}(1 - 1/P \mid \int_{\zeta} \zeta f_Z(\zeta) d\zeta) \\q_2 &= \mathbb{E}_Z[F_{A|Z}^{-1}(1 - 1/P \mid Z)] = \int_{\zeta} F_{A|Z}^{-1}(1 - 1/P \mid \zeta) f_Z(\zeta) d\zeta \\q_3 &= \tilde{F}_A^{-1}(1 - 1/P) \text{ where } \tilde{F}_A(x) = \int_{\zeta} F_{A|Z}(x \mid \zeta) f_Z(\zeta) d\zeta \\q_4 &= \tilde{F}_{A_P}^{-1}(\exp(-1)) \text{ where } \tilde{F}_{A_P}(x) = \tilde{F}_A^P(x) \\q_5 &= \text{med}_Z[F_{A|Z}^{-1}(1 - 1/P \mid Z)]\end{aligned}$$

- For inference from large samples, estimators provide similar estimates
- For small samples, estimates can be very different
 - Even when using the same statistical model, and identical data
- Be careful in comparing return value estimates from different sources

Return values with uncertainty



Devil in the detail: fractional bias from 5 estimators for return value.
Posterior predictive (q_3) over-estimates for small samples. Median (q_5) reasonable. See Towe et al. [2023]

Return values with uncertainty

References

- Jonathan et al. [2021] (for stationary marginal extremes)
- Towe et al. [2023] (for associated values from conditional extremes)
- Tendijck et al. [2024b] (for Shell CEVA software)

Wrap-up: cheat sheet

- Estimating tails of distributions is challenging, lots of approaches
 - If you're unsure what you're doing, then try multiple models
 - If inferences about the target diverge $\Rightarrow$ **RED FLAG**
- Doing inference on output of relevant (physical or statistical) simulation models is a rigorous way to evaluate different model choices
- Incorporate physical knowledge
 - Does a modelling assumption make sense from an engineering perspective?
 - Does a modelling assumption make sense statistically?
- Always perform sensitivity analysis with respect to model fiddle factors
 - If large variability in target inferences $\Rightarrow$ **RED FLAG**
 - If possible, use cross-validation to estimate optimal fiddle factor values
- Produce key diagnostic plots for any model you consider
- Incorporate uncertainty from statistical inference in decision making
 - Bootstrapping, model averaging all easy; no excuses
 - Bayesian methods provide consistent framework but steeper learning curve
- Essential to make analysis genuinely reproducible by others in future
 - Document reasoning, choices, assumptions, model performance
- Small samples require simple models
 - Good models default to simple ones when no evidence for complexity
- Full probabilistic design is the future

Wrap-up: cheat sheet

Covariates

- Directional models essential to estimate directional extremes consistently
- Non-stationary extreme value models can get complicated very quickly
 - But good modelling frameworks default to stationary models when no evidence for covariate effects
- Always benchmark against thoughtful alternatives (e.g. covXtreme)

Multivariate

- Applied multivariate extreme value analysis is less well developed
 - Lots of options involving strong assumptions (AD versus AI)
- Conditional extremes more general, simple, and used in other fields
- Emerging angular-radial models like SPAR (admitting AD and AI)

Wrap-up: best practice mantra?

Right balance of good engineering and good statistics
(Joined-up marginal and dependence modelling, aim for probabilistic design,
especially for high-risk applications)

Empirical inference demands thorough uncertainty assessment and
propagation throughout analysis procedure
(Cross-validation, bootstrapping, model averaging; or Bayesian inference)

Reproducible analysis as rational basis for learning
(Audit trail documenting key assumptions, equations and diagnostic results)

All models are wrong, some models are useful

Diolch & thank-you!

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